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# How to Pick the Model: A Framework for Budget-Aware Orchestration

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## Abstract

Agent systems on verified tasks invite a deceptively simple question: which backbone model should be run? The strongest model is not always the best deployment choice, because verified progress must be bought under finite wall-clock, token, API, and evaluation budgets. ~~We formulate this problem as deployment-aware model selection.~~ We formulate this problem as deployment-aware model selection for verified agent systems. A fixed router observes a pre-deployment information state and chooses a worker or harness action; the decision is scored by a checker-indexed deployment loss rather than by routing accuracy or final checker score alone. The key estimand is paired: on the same task instances, with the same router and action menu, does richer information change the selected action and reduce net deployment loss after its own measurement and context costs are charged? We instantiate the framework on a Karpathy AutoResearch-inspired CIFAR-10 edit-verify benchmark, where a prompt-only router chooses among heterogeneous workers before costly deployment runs. ~~The protocol reports when checker-score rankings disagree with deployment loss rankings, when pre-deployment information changes routing, and whether those changes pay for themselves.~~ The promoted two-worker study shows that first-hit speed, final improvement, log-effort, and deployment loss can rank workers differently. It also gives a negative routing result: probe signals expose the task modes, but the evaluated prompt-only router does not convert that information into positive paired deployment gain. ~~The resulting protocol evaluates not which agent is best in isolation, but when information is worth buying before deploying one.~~ The resulting protocol evaluates not which agent is best in isolation, but whether a router can buy and use information well enough to improve deployment value.

## 1 Introduction

Practitioners building agent systems face multiple practical challenges: ~~what model~~ which model (backbone) to use, how to route between models, when to spawn sub-agents, how to engineer context between them, and so on. All these possible choices often affect performance (quality of the solution), speed (time to solution), and ~~the mere token cost~~ token/API cost. The question is how to ~~pick between the different possible designs~~ choose among the possible designs. The common way to answer this question is benchmark racing: take a benchmark ~~as such as~~ SWE-bench [1] or MLE-bench [2] and pick the agent system with the best average end-to-end performance. An increasingly long line of work ~~, based on this “recipe”,~~ based on this recipe is proposing better and better ~~agentie system architecture~~ agent system architectures to achieve ~~SOTA at~~ state-of-the-art performance on these tasks. ~~This article investigates:~~ This article investigates

~~whether there exist other ways to answer the question of what the most suited agent system is:~~  
*whether there are other ways to answer the question of which agent system is most suitable.*

Indeed, (i) a large number of tasks for which such benchmark does not exist (as developing academic research [?]). And (ii) very often, the task presents instances of varied difficulty there are many tasks for which such benchmarks do not exist, such as developing academic research [3]. Moreover, tasks often contain instances of varied difficulty and the practitioner may want to understand whether, on the single instance on a single instance, a lower-cost, faster model is sufficient; more tools should be given to the system the system should be given more tools; or a more involved orchestration is required. As a running example for the rest of the paper, the question whether the user should use on their own task Haiku vs Sonnet vs Opus in Claude Code, or Claude Code vs Codex, the question of whether a user should use Haiku, Sonnet, or Opus in Claude Code, or Claude Code rather than Codex, for their own task is not fully addressed by benchmark racing.

The central issue is that one benchmark score may conflates conflate mechanisms that imply different engineering decisions. A system can win because its worker takes has lower per-step time or token count. Because the selected worker is token cost, or because the selected worker is more competent on a solver-relevant task mode. Analogously, imagine some sort of pre-deployment signal a pre-deployment signal (problem definition, the context, or an initial test the problem definition, context, or an initial test) reveals information about the instance which that can be used to better orchestrate or to choose what backbone to pick to choose a better backbone or orchestration. A system can win if it can orchestrate or choose the model optimally exploiting that information or not by exploiting that information when choosing a model or orchestration, or it can lose by failing to exploit it—as humans implicitly do when they decide to use lower or higher reasoning effort reasoning or different models on their tasks.

More practically, in this article we study the deployment problem of choosing which model or harness to run on a verifiable task. The objective is time to a required verified result time to reach a required verified result: among actions that may eventually succeed, the useful one is the one expected to reach the verifier threshold fastest, after charging any pre-deployment measurement used to make the choice. Achille and Soatto, indeed, recently showed that for verifiable tasks the time to solution, as in their Proper Time, is the performance measure to compare them. Indeed, Achille and Soatto [4] recently argued that time to solution, as formalized by their Proper Time, is a performance measure for comparing agents on verifiable tasks. Building on this intuition we operationalize the difference in time to solution of two possible designs. We show it mathematically decomposes in 4 components which isolates the mechanisms of choice mentioned above. Building on this intuition, we operationalize a certified log-effort surrogate gap between two possible designs. We show that it mathematically decomposes into four components that isolate the mechanisms of choice mentioned above. The experimental protocol then shows these can be estimated and lead to improvements in speed on Karpathy’s AutoResearch tasks. The experimental protocol estimates the operational quantities that are directly measurable and tests, by paired selected-worker deployment, whether richer pre-deployment information improves net deployment loss after measurement costs are charged.

**Research questions and contributions.** We organize the paper around three questions, each paired with the contribution that answers it.

1. Can the time-to-solution certified log-effort surrogate gap between agent designs be decomposed into meaningful terms? Theorem 1 gives an exact four-term identity under a packed latent-mode assumption: cost, within-mode competence, routing information, and a predeclared mode-summary divergence additively explain the improvement in expected certified log-effort.
2. Can the decomposition replace structure or reduce end-to-end racing for model selection? Algorithm 1 and the pilot concentration bound in Appendix A turn the identity into a structured pilot protocol whose evaluation cost depends on the number of models and task modes.
3. When does a lower-cost model with retries beat a stronger model? Theorem 2 gives the single-mode crossover rule, and the multi-mode accounting shows why the answer changes across task modes.

The experiments below instantiate these claims with checked trajectories and deployment accounting.

## 2 Problem-setup-and-deployment-estimands Setup and accounting

We study the problem faced by a deployment system before launching an agentic run: given a task instance and a finite menu of possible workers or harnesses, which one should be run? The answer is not necessarily the model with the best final score. A deployment action must produce checked progress under finite time, token, API, and evaluation budgets.

**Definition 1** (Checked deployment task). *A checked deployment task is a tuple*

$$\mathcal{T} = (\mathcal{X}, \mathcal{Y}, V, \mu),$$

where  $\mathcal{X}$  is an instance space,  $\mathcal{Y}$  is an artifact or solution space,  $V$  is an external checker or scoring procedure on instance–artifact pairs, and  $\mu$  is the deployment distribution over instances. For a realized instance  $x \in \mathcal{X}$ , an agent produces artifacts  $y \in \mathcal{Y}$  that are scored by  $V(x, y)$ .

The checker may be binary, as in unit tests or proof checking, or scalar, as in validation loss or reward. The key point is externality: progress is determined by the checker, not by the model’s self-report.

**Definition 2** (Actions and routers). *Let  $\mathcal{A}$  be a finite menu of deployable actions. An action may be a single model, a fixed model harness, a retry policy, or a tool-using agent protocol. For action  $a \in \mathcal{A}$  and instance  $x$ , let*

$$\ell_{\text{dep}}(a, x; \xi)$$

be the predeclared deployment loss of running action  $a$  on  $x$ , where  $\xi$  denotes deployment randomness such as sampling, system latency, or data-order randomness. The loss may include checked progress, failure, persistence above threshold, wall-clock time, token/API cost, and checker/evaluation cost.

A router  $R$  observes a permitted pre-deployment information state  $W(x)$  and selects an action

$$a_R(x) = R(W(x)) \in \mathcal{A}.$$

The selected action then performs the deployment run from the original task state.

**Definition 3** (Budgeted externally checked task). *A ~~checked deployment task~~ budgeted externally checked task is a tuple*

$$\mathcal{T} = (\mathcal{X}, \mathcal{Y}, V, B_{\text{dep}}, \mu).$$

Here  $\mathcal{X}$  is an instance space,  $\mathcal{Y}$  is a ~~artifact or solution space~~ solution space,  $V : \mathcal{X} \times \mathcal{Y} \rightarrow \mathcal{R}$  is an external checker or scoring procedure on instance–artifact pairs,  $B_{\text{dep}}$  is the full deployment budget and acceptance object, and  $\mu$  is the deployment distribution over instances. The object  $B_{\text{dep}}$  may include a horizon, verifier budget, token/time/cost caps, and a thresholding rule that induces a binary success event from the checker score. For a realized instance  $x \in \mathcal{X}$ , an agent produces artifacts  $y \in \mathcal{Y}$  that are scored by  $V(x, y)$ . An agent succeeds on  $x \sim \mu$  if it produces an accepted  $y$  before exhausting  $B_{\text{dep}}$ .

The checker may be binary, as in unit tests or proof checking, or scalar, as in validation loss or reward. The key point is externality: progress is determined by the checker, not by the model’s self-report.

**Definition 4** (Workers, actions, and routers). *Let  $\mathcal{M} = \{M_1, \dots, M_K\}$  be a finite menu of workers, where a worker is any deployable solver or fixed agent policy that can produce a candidate solution for verification. Let  $\mathcal{A}$  be a finite menu of deployable actions. An action may be a single model, a fixed model harness, a retry policy, or a tool-using agent protocol. In the simplest setting,  $\mathcal{A} = \mathcal{M}$ ; more generally, an action bundles a worker with fixed stopping, retry, tool-use, and carryover rules. ~~The deployment loss is defined locally in this action definition.~~ The deployment loss is defined in Section 4, where it is tied to first-passage cost, audit loss, and measurement loss. A routed design is a tuple*

$$d = (\mathcal{A}, R, \kappa),$$

where  $R$  is a router that selects an action after observing permitted information and  $\kappa$  is the declared scalar action-cost convention used in the log-effort diagnostic. When costs are logged as resource vectors, the protocol must fix the scalar reduction before evaluation. A router  $R$  observes a permitted pre-deployment information state  ~~$W(x)$~~   $Z_j(x)$  and selects an action

$$a_j(x) = R(Z_j(x)) \in \mathcal{A}.$$

Equivalently, a signal condition induces the routing policy  $\rho_j : x \mapsto a_j(x)$ . The selected action then performs the deployment run from the original task state.

This separates routing from deployment. The router chooses what to run; the selected action performs the actual checked run. Unless explicitly declared otherwise, pre-deployment records are shown only to the router and are not passed to the selected worker.

**Definition 5** (Task instances and task modes). Given  $\mathcal{T} = (\mathcal{X}, \mathcal{Y}, V, B_{\text{dep}}, \mu)$ , a task instance is a concrete realization  $X = x \in \mathcal{X}$  of that same task. It is the measurable object supplied to the deployed system; the router and worker may access only the fields made available by the protocol. A task mode is a latent variable

$$S = \sigma(X) \in \mathcal{S} = \{1, \dots, m_S\}$$

induced by a measurable map  $\sigma : \mathcal{X} \rightarrow \mathcal{S}$  that assigns each instance to one of finitely many solver-relevant variants of the task. It defines a prior  $\pi_s = \mathbb{P}_\mu[S = s]$  and mode-conditional instance distributions  $\mu_s = \mu(\cdot \mid S = s)$ . Modes are therefore not different tasks and not worker labels. They are variants of the same task: finite subpopulations of instances that share solver-relevant structure and may favor different workers or interventions.

~~The mode variable is retained by the evaluator for accounting and diagnostic comparisons. It need not be observed by either the router or the worker. This lets the same theory cover expert-defined task-mode schemas and finite evaluator-defined modes. When an experiment uses one fixed executable prototype per mode, it estimates conditional panel quantities rather than the population quantities induced by  $\mu$ . The evaluator retains  $S$  only for accounting: neither the router nor the worker needs to observe it. In the experiments, each mode is represented by one fixed executable prototype, so the reported quantities are fixed-panel analogues rather than population-level estimates under  $\mu$ .~~

**Assumption 1** (Packed latent-mode family). Each instance  $X \sim \mu$  is associated with a mode  $S = \sigma(X)$  in a finite mode set  $\mathcal{S}$ , with prior  $\pi$ . Conditional on mode  $s$ , action  $a$  has success probability  $p(a, s) > 0$  and scalar cost

$$\kappa(a, s) = \mathbb{E}[C_\delta(a, X; \xi) \mid S = s] > 0$$

under the declared deployment protocol, where  $C_\delta$  is the normalized cost accumulated to the first certified success or the horizon. At deployment time, the router observes a signal  $Z = z$  and selects one action. The evaluator defines the mode distribution induced by that signal,

$$\pi_z(s) = \mathbb{P}[S = s \mid Z = z].$$

The router is not required to output  $\pi_z$  or any mode distribution. Instead, the theory uses a fixed evaluator-side mode-summary distribution  $q_z \in \Delta(\mathcal{S})$ , with  $q_z(s) > 0$  whenever  $\pi_z(s) > 0$ . This summary is not a router output; it is a theory-facing diagnostic distribution specified before evaluation. Write  $a_z = \rho(z)$  for the action selected after observing  $z$ . For an instance whose true mode is  $s$ , the certified effort surrogate is proportional to

$$T_{\rho, q}(s, z) \propto \frac{\kappa(a_z, s)}{p(a_z, s) q_z(s)}. \quad (1)$$

~~Assumption 1 is deliberately restrictive. Real task instances can vary in infinitely many ways, while the theory uses a finite state abstraction. The assumption says that, for model selection, instances can be packed into modes with stable action success probabilities. The purpose is to isolate the regime in which cost, competence, information, and the predeclared mode-summary divergence are algebraically separable. Residual and calibration diagnostics measure when this finite packing is too coarse. Assumption 1 is a diagnostic abstraction, not an empirical claim that the finite schema is universally correct. It isolates the regime where cost, competence, signal information, and mode-summary mismatch are algebraically separable; residual, calibration, and held-out deployment diagnostics test whether this abstraction is useful for the declared panel.~~

**Definition 6** (Cost-adjusted certified log-effort diagnostic objective). For a fixed routed policy  $\rho : Z \mapsto a_Z$  and a fixed evaluator summary  $q$  satisfying Assumption 1, define

$$\mathcal{J}(\rho, q) = \mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} [\log \kappa(a_Z, S) - \log p(a_Z, S)] + \mathbb{E}_Z [\mathcal{H}(\pi_Z) + \text{KL}(\pi_Z \| q_Z)]. \quad (2)$$

Lower  $\mathcal{J}$  means lower expected certified log-effort under the fixed diagnostic convention. The quantity  $q$  is not optimized by the router; it is fixed by the evaluator-side diagnostic rule. When several signal conditions are compared,  $\mathcal{J}$  is applied to the corresponding policy summaries; Appendix C writes this comparison explicitly for  $Z_j$  versus  $Z_0$ .

The action quantities in (2) are  $\kappa(a_Z, s)$  and  $p(a_Z, s)$ : how costly the selected action is on mode  $s$  and how often it succeeds there. The router and information quantities are  $Z$ ,  $\pi_z$ , and  $q_z$ : what the router is allowed to observe, what that signal reveals about the task mode, and the evaluator-side mode-summary distribution used in the KL term.

**Remark 1** (Meaning of the effort surrogate). Equation (1) says that, on a true mode  $s$  after signal  $z$ , certified effort increases with action cost, decreases with action success probability on that mode, and increases when the evaluator-side mode summary gives too little diagnostic mass to that mode. The router is not literally routing jobs to modes; modes are evaluator-defined task variants. The quantity  $q_z(s)$  is a diagnostic summary, not a deployment probability. The KL term below is therefore a mode-summary divergence under the chosen diagnostic convention. It should be interpreted as a router failure only if the experiment predefines a reproducible construction of  $q_z$  from router behavior.

### 3 Performance-accounting Accounting identity

For a fixed signal value  $z$ , the routing part of the objective is the cross-entropy between the signal-induced mode distribution and the fixed evaluator-side mode-summary distribution:

$$\text{CE}(\pi_z, q_z) = - \sum_{s \in \mathcal{S}} \pi_z(s) \log q_z(s) = H(\pi_z) + \text{KL}(\pi_z \| q_z). \quad (3)$$

The KL term is a divergence over finite task-mode distributions. It is not a token-level language-model divergence.

**Theorem 1** (Four-term routed-policy decomposition). Under Assumption 1, compare a baseline policy  $\rho_0$  that deploys fixed action  $a_0$  without using  $Z$  and uses  $q_0(\cdot | z) \equiv \pi$ , with a deployed routed policy summarized by  $(\rho, q)$ . The improvement in expected certified log-effort,  $\Delta = \mathcal{J}(\rho_0, q_0) - \mathcal{J}(\rho, q)$ , decomposes as

$$\Delta = \underbrace{\mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} \left[ \log \frac{\kappa(a_0, S)}{\kappa(a_Z, S)} \right]}_{\text{cost}} + \underbrace{\mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} \left[ \log \frac{p(a_Z, S)}{p(a_0, S)} \right]}_{\text{competence}} + \underbrace{\mathbb{I}_\pi(S; Z)}_{\text{routing information}} - \underbrace{\mathbb{E}_Z [\text{KL}(\pi_Z \| q_Z)]}_{\text{mode-summary divergence}}. \quad (4)$$

*Proof sketch.* Taking logs in (1) gives  $\log T_{\rho, q}(s, z) = C + \log \kappa(a_z, s) - \log p(a_z, s) - \log q_z(s)$ . Conditioning on  $Z = z$  and averaging over  $S \sim \pi_z$  converts  $-\mathbb{E} \log q_z(S)$  into the cross-entropy in (3). Subtracting the prior-matched baseline cancels constants and uses  $H(\pi) - \mathbb{E}_Z H(\pi_Z) = \mathbb{I}_\pi(S; Z)$ . The full algebra is given in Appendix A.  $\square$

The four terms have different operational meanings. If the cost term is large and positive, a lower-cost action may be preferable even when it reaches success later in step count. If the competence term varies by mode, a global worker ranking is likely to hide comparative advantage. If information gain is high, the pre-deployment signal exposes solver-relevant structure. If the mode-summary divergence is high, the signal-induced mode distribution is poorly matched to the evaluator’s predeclared mode summary. For two signal conditions, the identity compares the routed policies induced by those signals. A richer signal is useful only if its induced action change improves the cost–success tradeoff after the cost of obtaining the signal is charged.

**Operational reading.** The task mode  $S$  is not an oracle label for the best worker. The evaluator fixes a mode schema before evaluation, then estimates the mode-conditional quantities  $p(a, s)$  and  $\kappa(a, s)$ . Routing information asks whether a signal makes those modes more identifiable. Routing shift asks whether the selected action changes when the signal is enriched. Mode-summary divergence is the KL term defined by the fixed evaluator-side mode summary  $q$ . Allocation regret, used later in the empirical protocol, is a separate action-frontier diagnostic rather than an estimator of this KL term. The final decision is not made by the information term alone; it is made by paired deployment gain on held-out instances after measurement and context costs are charged. Appendix B gives the full operationalization of these diagnostics.



### 3.1 ~~When lower-cost retries win~~Retry crossover design rule

**Theorem 2** (Single-mode retry crossover). *Fix a single homogeneous mode, success threshold  $\delta$ , and horizon  $H$ . Let actions  $h$  and  $\ell$  have one-attempt first-passage success probabilities  $p_h = \mathbb{P}[\tau_\delta(h, X) \leq H]$  and  $p_\ell = \mathbb{P}[\tau_\delta(\ell, X) \leq H]$ , with expected first-passage costs  $\kappa_h$  and  $\kappa_\ell$ . Assume  $0 < p_\ell < p_h < 1$  and  $\kappa_\ell < \kappa_h$ . Assume retries of  $\ell$  are conditionally i.i.d. fresh attempts, each with the same one-attempt horizon  $H$  and common Bernoulli success probability  $p_\ell$ . The lower-cost retry depth that matches the strong action's one-attempt success probability is*

$$D_{\text{cross}} = \frac{\log(1 - p_h)}{\log(1 - p_\ell)}.$$

The corresponding integer fixed-depth retry block is cost-favorable whenever

$$\lceil D_{\text{cross}} \rceil \kappa_\ell < \kappa_h.$$

**Remark 2** (Stopping after first retry success). *If retries of the lower-cost action stop after the first success and each attempted retry has unconditional expected cost  $\kappa_\ell$ , the expected cost at depth  $D$  is*

$$\kappa_\ell \sum_{d=1}^D (1 - p_\ell)^{d-1} = \kappa_\ell \frac{1 - (1 - p_\ell)^D}{p_\ell},$$

and the decision should also account for the residual failure probability under the deployment loss. Any difference in within-attempt time-to-success is already absorbed into  $\kappa_h$  and  $\kappa_\ell$ .

The multi-mode case is where the four-term identity matters. A lower-cost worker can be rational on one task mode and irrational on another. A router is useful only when different actions win the cost–success tradeoff on different modes and the pre-deployment information helps identify which mode applies.

## 4 ~~Decomposition-guided selection~~Deployment validation protocol

~~The decomposition suggests a practical protocol. Rather than evaluate every complete design as an independent arm, the builder first estimates shared quantities: mode-conditional success, cost, predeclared signal diagnostics, and action-allocation diagnostics. Then a larger finite design menu can be scored analytically, with expensive deployment evaluations reserved for finalists and for paired validation. The log-effort decomposition guides diagnostics, but deployment recommendations are made only after paired validation under a predeclared loss. The empirical protocol is compact: freeze modes, losses, cost normalizers, and signal records; estimate mode-conditional worker success and first-passage cost; audit whether richer records expose modes and change allocation; then validate finalists by paired selected-worker deployment after measurement cost.~~

The log-effort objective  $\mathcal{J}$  is a diagnostic surrogate, not the deployment loss.

**Diagnostic versus decision loss.**  $\mathcal{J}$  explains cost–competence and signal/accounting structure under a declared diagnostic convention.  $\ell_{\text{dep}}$  is the decision-facing loss used for paired deployment validation. We use  $\mathcal{J}$  to diagnose why rankings differ, and  $\ell_{\text{dep}}$  to decide whether a routed policy actually improves deployment value.

For deployment validation we use a predeclared first-passage loss. Given a checked progress trace  $G_1, \dots, G_H$ , define

$$\tau_\delta(a, x; \xi) = \inf\{t \leq H : G_t(a, x; \xi) \geq \delta\}, \quad F_\delta(a, x; \xi) = \mathbf{1}\{\tau_\delta(a, x; \xi) = \infty\},$$

and

$$\text{Occ}_{\delta, H}(a, x; \xi) = H^{-1} \sum_{t=1}^H \mathbf{1}\{G_t(a, x; \xi) \geq \delta\}.$$

Let  $c_\delta(a, x; \xi)$  be the normalized worker-side resource vector accumulated to  $\tau_\delta \wedge H$ , and let

$$C_\delta(a, x; \xi) = \lambda^\top c_\delta(a, x; \xi)$$

251 be the scalar cost used in  $\kappa(a, s)$ . The primary loss is

$$\ell_{\text{dep}}(a, x; \xi) = \alpha_{\text{fail}} F_{\delta}(a, x; \xi) + C_{\delta}(a, x; \xi), \quad (5)$$

252 with  $\alpha_{\text{fail}} = 1$  and normalized resource weights  $\lambda$  fixed before evaluation. This is the loss used  
 253 in  $\mathcal{J}_R(j)$  and  $\Delta_R(j)$  unless a different primary loss is explicitly declared. Because the benchmark  
 254 records full trajectories to horizon  $H$ , we also report the full-horizon audit loss

$$\ell_{\text{audit}}(a, x; \xi) = \ell_{\text{dep}}(a, x; \xi) + \alpha_{\text{occ}}(1 - \text{Occ}_{\delta, H}(a, x; \xi)) + \alpha_{\text{qual}}\psi(G_H(a, x; \xi)), \quad (6)$$

255 where  $\alpha_{\text{occ}} = \alpha_{\text{qual}} = 1$  and  $\psi(u) = 1 - u$  after clipping  $G_H \in [0, 1]$ . The audit loss checks  
 256 persistence and final quality; it does not change the first-passage cost convention in Theorem 1.

257 **Algorithm 1** (Theory-to-deployment selection protocol).

*Input:* Task family, checker, finite task-mode schema  $\mathcal{S}$ , action menu  $\mathcal{A}$ , admissible progressive signals  $\{Z_j\}$ , pilot cell count  $n_{\text{cell}}$ , deployment repeats  $Q_{\text{dep}}$ , confirmation budget.

*Output:* Recommended signal-conditioned deployment policy, reported accounting terms, paired deployment gain, and calibration diagnostics.

- 1 *Freeze the modes and loss.* Declare  $\mathcal{S}$ , success threshold  $\delta$ , horizon  $H$ , cost normalizers, deployment loss (5), measurement costs, and router output contract.
- 2 *Structured pilot.* For each action  $a \in \mathcal{A}$  and mode  $s \in \mathcal{S}$ , estimate  $\hat{p}(a, s)$ ,  $\hat{\kappa}(a, s)$ , first-hit curves, occupancy, and uncertainty intervals.
- 258 3 *Signal and router diagnostics.* For each progressive signal  $Z_j$ , report any predeclared signal-information diagnostic and log the action allocation induced by the router.
- 4 *Accounting score.* Compute the four log-effort terms from Theorem 1 only for quantities whose diagnostic estimators were predeclared; otherwise report allocation regret separately.
- 5 *Paired deployment validation.* On the same holdout instances, run the selected action under  $Z_0$  and richer  $Z_j$  conditions, score repeated selected-worker trajectories by (5), and subtract measurement cost.
- 6 *Decision.* Choose the signal condition with the largest held-out  $\hat{\Delta}_R(j)$  subject to predeclared gates; use  $\mathcal{J}$ , frontier scores, and allocation regret as diagnostics, not as the final deployment objective.

#### 259 4.1 Progressive signals and paired router value

260 Let  $Z_j(X; \zeta_j)$  be the router-visible pre-deployment record under signal condition  $j$ , where  $\zeta_j$  denotes  
 261 measurement randomness. Let  $Z_0$  be the static signal observed before choosing a deployment action.  
 262 For any admissible richer condition  $j$ , the router observes  $Z_j$ , pays the corresponding measurement  
 263 and router-context cost, and selects an action

$$a_R^j(X) = R(Z_j(X; \zeta_j); U_R) \in \mathcal{A}.$$

264 Here  $U_R$  denotes router-side randomness; for deterministic routers it is fixed and suppressed. Thus  
 265 the experimental comparison is between routing policies  $\rho_0, \rho_j$  induced by different signals for the  
 266 same orchestrator  $R$ , not between different orchestrators. The normalized measurement loss is

$$\ell_{\text{meas}}(j, x; \zeta_j) = \lambda_{\text{meas}}^{\top} d_j(x; \zeta_j),$$

267 where  $d_j$  contains measurement wall-clock, tokens, checker calls, context expansion, parsing, retry,  
 268 and latency costs. Set  $\ell_{\text{meas}}(0, x) = 0$ . The weights  $\lambda_{\text{meas}}$  use the same predeclared normalizers  
 269 as deployment resource cost unless a separate router-cost scale is declared. In the main protocol,  
 270 pre-deployment probes are outside the selected action's deployment run: they are charged through  
 271  $\ell_{\text{meas}}$  and do not reduce the later horizon  $H$  or verifier budget  $B_V$ .

272 The selected-action deployment objective is

$$\mathcal{J}_R(j) := \mathbb{E} \left[ \ell_{\text{meas}}(j, X; \zeta_j) + \ell_{\text{dep}}(a_R^j(X), X; \xi) \right]. \quad (7)$$

273 The expectation averages over task instances, measurement randomness, router randomness, and  
 274 deployment randomness unless a conditional estimand is explicitly declared. The router-facing value  
 275 of signal condition  $j$  is the paired gain

$$\Delta_R(j) := \mathcal{J}_R(0) - \mathcal{J}_R(j) = \mathbb{E} \left[ \ell_{\text{dep}}(a_R^0(X), X) - \ell_{\text{dep}}(a_R^j(X), X) - \ell_{\text{meas}}(j, X) \right]. \quad (8)$$

276 Positive  $\Delta_R(j)$  means that the richer signal improves net deployment loss for the same router on the  
 277 same task distribution. Allocation shift alone is insufficient:

$$\text{Shift}_R(j) = \Pr[a_R^j(X) \neq a_R^0(X)]$$

278 measures decision relevance, not benefit.

279 **Proposition 1** (Unbiased paired router-gain estimator). *Assume task instances are sampled i.i.d. from*  
 280  *$\mu$ , the orchestrator, action menu, and signal protocols are fixed before evaluation, measurement costs*  
 281 *are recorded without bias, and deployment seeds are independent conditional on selected action and*  
 282 *instance. The router is run once per instance and signal condition; the  $Q_{\text{dep}}$  deployment repeats*  
 283 *estimate the conditional expected loss of the selected action. For condition  $j$ , estimate*

$$\hat{\Delta}_R(j) = \frac{1}{N} \sum_{i=1}^N \left[ \frac{1}{Q_{\text{dep}}} \sum_{q=1}^{Q_{\text{dep}}} \ell_{\text{dep}}(a_{R,i}^0, x_i; \xi_{i,q}^0) - \frac{1}{Q_{\text{dep}}} \sum_{q=1}^{Q_{\text{dep}}} \ell_{\text{dep}}(a_{R,i}^j, x_i; \xi_{i,q}^j) - \hat{\ell}_{\text{meas}}(j, x_i) \right].$$

284 *If the repeated-run averages are unbiased for the expected deployment losses of the selected actions,*  
 285 *then  $\mathbb{E}[\hat{\Delta}_R(j)] = \Delta_R(j)$ .*

286 When the same worker is selected under  $Z_0$  and  $Z_j$ , using common deployment seeds preserves  
 287 unbiasedness and reduces paired variance. When counterfactual outcomes for all actions are logged  
 288 on the same instance, hindsight action regret can be reported as a diagnostic, but it is not the primary  
 289 estimand. If the study fixes a finite prototype panel rather than sampling i.i.d. instances from  $\mu$ , the  
 290 same estimator describes that panel and should not be interpreted as a population-level deployment  
 291 estimand.

## 292 5 Experiments and analysis

293 **Setup.** We instantiate the framework on a fixed-prototype repeated-trajectory panel for CIFAR-10  
 294 edit-verify optimization. The task modes are three starting optimization variants of the same checked  
 295 task:

$$\mathcal{S} = \{\text{MLP-flat}, \text{compact CNN}, \text{micro ResNet}\},$$

296 with empirical weights  $\hat{\pi}(s) = 1/3$ . They are task modes because the dataset, checker, horizon, and  
 297 success rule are held fixed, while the starting architecture family and parameterization of the editable  
 298 artifact change. This creates different optimization landscapes and potentially different worker  
 299 cost-success frontiers. We use these modes to test whether worker choice is mode-conditional rather  
 300 than a single global ranking. The analyzed worker menu is  $\{\text{gpt\_5\_3\_codex}, \text{gpt\_5\_4}\}$ . Each  
 301 action is one worker under the common prompt and harness, so  $\mathcal{A} = \mathcal{M}$  in this experiment. For each  
 302 task-mode-worker cell we use 35 trajectories. The router is a deterministic prompt-only orchestrator  
 303 evaluated under four nested pre-deployment records  $Z_0, \dots, Z_3$ , from budget-only context to full  
 304 unmodified-artifact baseline. The primary decision metric is the first-passage deployment loss  
 305 in (5); full-horizon occupancy and final quality are audit metrics. Appendix D gives the frozen  
 306 configuration, including horizon, checker budget, training budget, success threshold, signal contents,  
 307 measurement-cost convention, progress normalization, smoothing rule, and logging schema.

308 **Worker frontier.** Experiment 1 runs every worker on every task mode before routing, estimating  
 309 fixed-prototype analogues of  $p(a, s)$  and  $\kappa(a, s)$ . These estimates instantiate the worker-side cost and  
 310 competence terms in Theorem 1; the smoothed log-effort estimator is defined in Appendix D.

311 The frontier already falsifies a single-score recommendation. Ranking by final improvement or  
 312 first-hit speed makes  $\text{gpt\_5\_4}$  look dominant, but ranking by the charged deployment loss changes  
 313 the recommendation. This is the empirical role of the decomposition: it exposes whether apparent  
 314 performance comes from first-passage competence, cost, or mode-specific comparative advantage.  
 315 Appendix E provides the threshold sensitivity, trajectory-level progress, cost, token, and router-record  
 316 diagnostics behind these aggregate summaries.

317 **Signal and routing diagnostics.** Experiment 2 audits whether richer pre-deployment records  
 318 change the deterministic router’s decision state. This is not an estimator of  $I(S; Z_j)$ : a Shannon-  
 319 information estimate would require a frozen finite representation or predictive model for the signal  
 320 records. In the current diagnostic, budget-only records recover the three task-mode labels at 0.349



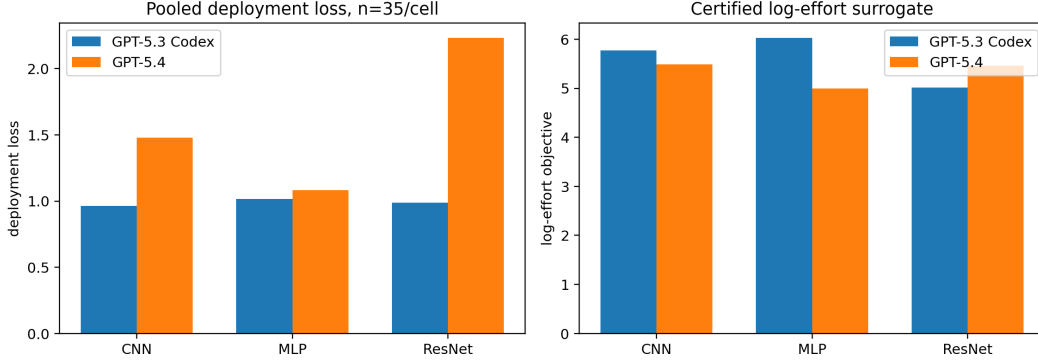


Figure 1: Two-worker frontier on the 35-run pooled panel. Left: first-passage deployment loss, lower is better. Right: certified log-effort surrogate. The two summaries disagree by mode: gpt\_5\_4 reaches the threshold faster and often has higher final checked improvement, while deployment-loss accounting favors gpt\_5\_3\_codex on *compact CNN* and *micro ResNet*.

Table 1: Worker-frontier summary on the 35-run pooled panel. Each cell uses 35 trajectories. Lower is better for log-effort and deployment loss.

| Mode                | Worker        | Success | Mean $\tau$ | Log-effort | Dep. loss |
|---------------------|---------------|---------|-------------|------------|-----------|
| <i>MLP-flat</i>     | gpt_5_3_codex | 32/35   | 7.41        | 6.029      | 1.015     |
| <i>MLP-flat</i>     | gpt_5_4       | 35/35   | 2.49        | 4.991      | 1.083     |
| <i>compact CNN</i>  | gpt_5_3_codex | 34/35   | 6.03        | 5.774      | 0.964     |
| <i>compact CNN</i>  | gpt_5_4       | 35/35   | 3.06        | 5.492      | 1.479     |
| <i>micro ResNet</i> | gpt_5_3_codex | 35/35   | 2.60        | 5.015      | 0.987     |
| <i>micro ResNet</i> | gpt_5_4       | 35/35   | 1.80        | 5.461      | 2.234     |

321 accuracy, while probe-containing records recover them perfectly on the diagnostic set. Thus the  
 322 signals can expose mode structure; the remaining question is whether the router converts that structure  
 323 into better actions.

324 Experiment 3 measures this conversion with shift rate and action-frontier regret. Shift rate is the  
 325 fraction of modes for which the richer signal changes the selected worker; action-frontier regret  
 326 scores the selected action against the mode-conditional frontier from Experiment 1. Both diagnostics  
 327 are defined in Appendix D.

328 The final audit contains 480 prompt-only decisions. On real signal records, the router remains strongly  
 329 biased toward gpt\_5\_4. Mean log-effort regret is 0.177 for  $Z_0$ , 0.183 for  $Z_1$ , 0.148 for  $Z_2$ , and 0.192  
 330 for  $Z_3$ . The non-deployable oracle in Figure 2 shows that the worker frontier contains exploitable  
 331 mode-conditional structure, but the evaluated router does not close that gap. Additional router-record  
 332 completion and signal-ablation plots are reported in Appendix E.

333 **Paired deployment gain.** Experiment 4 is the deployment-facing test. For each richer signal  $Z_j$ ,  
 334 we compare the action selected under  $Z_j$  with the action selected under  $Z_0$  on the same prototype  
 335 panel and subtract measurement cost:

$$\hat{\Delta}_R(j) = \hat{\ell}_{\text{dep}}(\rho_0) - \hat{\ell}_{\text{dep}}(\rho_j) - \hat{\ell}_{\text{meas}}(j).$$

336 Positive values mean that the richer signal improves net deployment loss after the cost of measuring  
 337 or using the signal is charged. This is the primary empirical estimand; positive signal information or  
 338 lower regret alone is insufficient.

339 The paired validation is negative for the current prompt-only router. Against  $Z_0$ , richer real signals  
 340 produce small shift rates and negative net gain after measurement cost. The bootstrap intervals from  
 341 the accounting script are  $[-0.116, 0.002]$  for  $Z_1$ ,  $[-0.127, -0.007]$  for  $Z_2$ , and  $[-0.113, -0.024]$

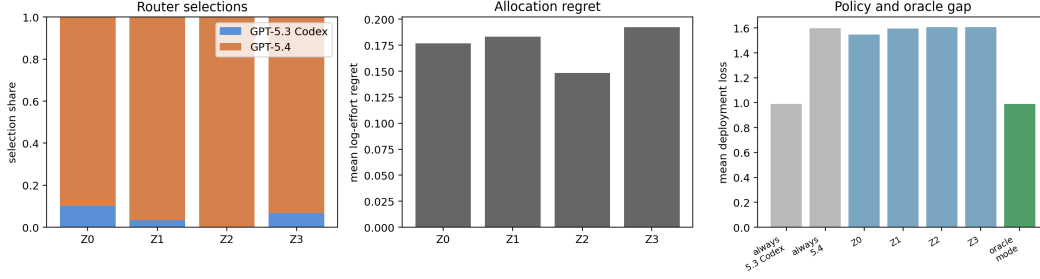


Figure 2: Two-worker router-response audit. Left: selected-worker distribution under real signal records. Middle: action-frontier regret against the 35-run pooled log-effort frontier. Right: deployment loss for always-worker baselines, prompt-router policies, and the non-deployable mode oracle. Richer records expose mode information but do not reliably move the prompt-only router toward the cost-adjusted frontier.

Table 2: Paired router validation on the final two-worker menu. Net gain is  $\hat{\ell}_{\text{dep}}(\rho_0) - \hat{\ell}_{\text{dep}}(\rho_j) - \hat{\ell}_{\text{meas}}(j)$ ; positive is better.

| Signal | Pairs | Shift | Gross gain | Meas. loss | Net gain |
|--------|-------|-------|------------|------------|----------|
| $Z_1$  | 30    | 0.133 | -0.049     | 0.000      | -0.049   |
| $Z_2$  | 30    | 0.100 | -0.051     | 0.007      | -0.059   |
| $Z_3$  | 30    | 0.100 | -0.032     | 0.028      | -0.060   |

for  $Z_3$ . This is the main falsification result: informative records are not enough; the deployed router must use them in a way that improves the selected action’s checked cost–success tradeoff.

**Negative controls and interpretation.** Experiment 5 repeats the router and paired-gain analyses with schema-preserving controls: shuffled probe records, wrong-mode probes, and synthetic noise records with matched field names and approximate context burden. Controls are charged under the same measurement/context-cost convention as the corresponding real signal. They test whether router response and paired gain are specific to task-relevant signal content.

The controls block a strong routing-information claim. Wrong-mode  $Z_2$  has a small positive mean net gain with an interval crossing zero, while several shuffled or synthetic controls are negative with intervals similar to the real signals. The correct conclusion is diagnostic rather than prescriptive: the worker frontier contains mode-conditional structure, and probes reveal that structure, but this router does not reliably turn the information into better deployment decisions. Appendix D states the full promotion rule used to separate diagnostic improvements from a deployable routing recommendation.

## 6 Related work

The closest classical ancestor is algorithm selection. Since Rice’s formulation, solver portfolios have treated performance as instance-dependent: SATzilla and AutoFolio learn to select solvers from instance features rather than report one globally best solver [5–8]. Our contribution is the specialization to externally checked LLM agents, where a router observes a controlled pre-deployment record and is scored by paired deployment loss rather than standalone routing accuracy. Unlike AutoML systems such as auto-sklearn, which tune pipelines offline for a dataset, our unit is a deployable action for a concrete verifier-backed instance [9].

LLM routing and cascades provide the model-menu context. FrugalGPT, RouteLLM, RouterBench, LLMRouterBench, and compound-system routers study cost-quality routing over heterogeneous model pools [10–16]. We ask how much checked, task-specific information is worth before committing to an expensive agentic deployment.

Checked agent benchmarks provide the task substrate. SWE-bench evaluates repository repair against tests, AutoResearch evaluates training-loop edits against validation feedback, and theorem-

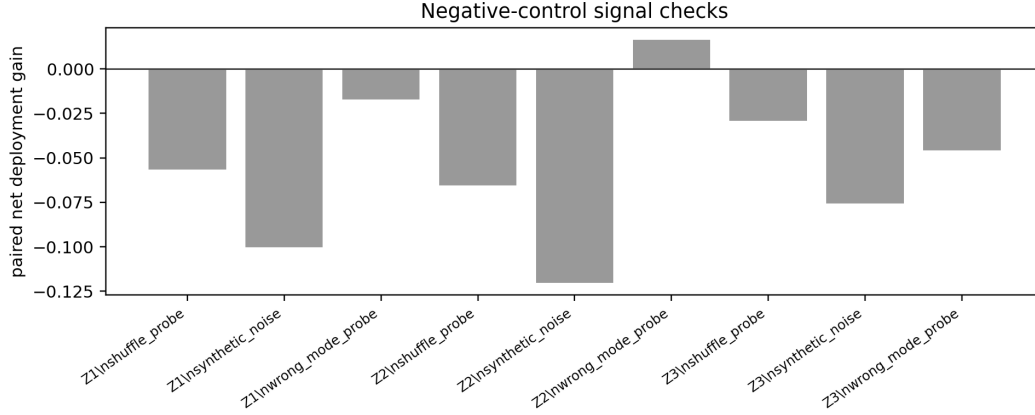


Figure 3: Negative-control paired gains for shuffled probes, wrong-mode probes, and synthetic noise records. The real records do not cleanly outperform the controls, indicating that the current prompt-only router is not reliably exploiting task-relevant measurement content.

369 proving systems use analogous external validators [1, 17]. Repeated sampling and inference-time  
 370 scaling further show that lower-cost or smaller models can become competitive when verification is  
 371 available [18–21]. Our question is upstream of the launched worker: which action should be selected,  
 372 what should the router observe, and how should that observation be charged?

## 373 7 Discussion and limitations

374 The paper has two distinct claims: an exact theory-facing accounting identity for a certified log-effort  
 375 surrogate, and a deployment-facing empirical claim that must pass paired-loss validation. There are  
 376 two levels of claim. The theory-facing claim is an accounting identity under Assumption 1. It is  
 377 exact, but only for the certified log-effort surrogate. The deployment-facing claim is empirical: the  
 378 accounting should predict or explain useful action selection under a richer deployment loss. These  
 379 two claims should not be collapsed. The current experiments support the central diagnostic claim:  
 380 raw worker rankings, first-passage rankings, log-effort rankings, and deployment-loss rankings can  
 381 disagree on the same checked task family. That disagreement is precisely the regime in which a  
 382 budget-aware orchestration framework is needed. They do not support the stronger claim that the  
 383 current prompt-only router improves deployment loss. Instead, the router validation is a useful  
 384 negative result: task-relevant information is measurable, but a router that mostly defaults to the  
 385 stronger model can still lose after measurement cost and worker-side costs are charged.

386 The packed-mode assumption is the main modeling limitation. Real agent trajectories are stateful,  
 387 attempts are not independent, and task modes may be fuzzy. The paper therefore reports first-passage  
 388 curves, occupancy, hazard diagnostics, cost sensitivity, and surrogate-versus-loss ranking agreement  
 389 instead of assuming that the finite-mode surrogate is correct. If these diagnostics are unstable, the  
 390 decomposition remains a useful diagnostic language but not a reliable selection rule.

391 The operational task-mode schema is also a limitation. If  $S$  is too coarse, mode-conditional compe-  
 392 tence is washed out and the router appears useless. If  $S$  is too fine, sample sizes per cell collapse and  
 393 the signal-mode estimates become unstable. The present evaluation fixes the executable prototypes  
 394 and mode schema to isolate agentic trajectory variance; task-instance and schema perturbations are  
 395 left as robustness extensions rather than promoted experiments.

396 Finally, measurement value is router-specific. A baseline probe can contain information about the  
 397 best action and still fail to improve deployment loss if the actual router does not use it well or if  
 398 the measurement is too expensive. For this reason, the paired gain  $\Delta_R(j)$  is the main deployment  
 399 estimand, while mutual-information diagnostics, mode-summary divergence, and allocation regret  
 400 help explain why that gain appears or fails to appear. This distinction is the main empirical lesson of  
 401 the final two-worker study. The  $Z_2$  and  $Z_3$  records contain task-mode information, but the induced

402 policies have negative paired gain. The decomposition therefore functions as a claim discipline: it  
403 prevents us from promoting information availability as deployment value.

404 **Future work.** The immediate experimental extension is to add a stable lower-cost action with  
405 matched trajectory support. That would make the worker menu span a broader cost–capability frontier  
406 than the two-worker comparison reported here. It would also test the retry-crossover theorem in  
407 its intended regime, where a lower-cost worker may compensate for lower one-attempt success  
408 through repeated checked attempts. Beyond this benchmark, the same protocol should be tested on  
409 repository repair, theorem checking, and multi-file research-code editing, where the mode schema is  
410 less artificial and measurement costs are more heterogeneous.

411 **Pier:** *qui in limitations very extensive! Dobbiamo pensarle tutte. Dire perché, argue that also related*  
412 *work has them. And then in future direction say you want to understand if you could work them*  
413 *around.*

414 **Conclusion.** Model choice inside an agent harness is not a scalar leaderboard problem. It is a  
415 measurable decomposition problem validated by paired deployment outcomes. The right worker  
416 depends on task state, checked competence, routing information, allocation quality, first-passage time,  
417 retry depth, and worker-side cost. Our CIFAR-10 edit–verify campaign provides the first empirical  
418 slice of that argument: the same two mature workers can trade off speed, final improvement, log-effort,  
419 and deployment loss in different ways. The two-worker routing analysis asks the deployment-facing  
420 question directly and answers it negatively for the current prompt-only router: pre-deployment  
421 information does not move allocation reliably enough to improve paired deployment loss after its  
422 measurement cost is charged. That negative result is not a failure of the framework; it is the framework  
423 doing its job, distinguishing worker-frontier structure from a validated routing recommendation.

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## A Proof details Proofs and retry rule

### A.1 Four-term identity

Under Assumption 1,

$$\log T_{\rho,q}(S, Z) = C + \log \kappa(a_Z, S) - \log p(a_Z, S) - \log q_Z(S).$$

Conditioning on  $Z = z$ ,

$$\mathbb{E}[-\log q_Z(S) \mid Z = z] = \sum_s \pi_Z(s) \log \frac{1}{q_Z(s)} = H(\pi_Z) + \text{KL}(\pi_Z \parallel q_Z).$$

Therefore

$$\mathbb{E}[\log T_{\rho,q}] = C + \mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} [\log \kappa(a_Z, S) - \log p(a_Z, S)] + \mathbb{E}_Z [H(\pi_Z)] + \mathbb{E}_Z [\text{KL}(\pi_Z \parallel q_Z)].$$

For the prior-matched baseline  $\rho_0$  that deploys fixed action  $a_0$  and uses  $q_0(\cdot \mid z) \equiv \pi$ ,

$$\mathbb{E}[\log T_{\rho_0,\pi}] = C + \mathbb{E}_{S \sim \pi} [\log \kappa(a_0, S) - \log p(a_0, S)] + H(\pi).$$

Subtracting the deployed effort from the baseline effort yields

$$\Delta = \mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} \left[ \log \frac{\kappa(a_0, S)}{\kappa(a_Z, S)} \right] + \mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} \left[ \log \frac{p(a_Z, S)}{p(a_0, S)} \right] + H(\pi) - \mathbb{E}_Z [H(\pi_Z)] - \mathbb{E}_Z [\text{KL}(\pi_Z \parallel q_Z)].$$

The entropy difference is  $I_\pi(S; Z)$ , giving (4).

## 489 A.2 Pilot concentration

490 For binary success observations in an action-mode cell, Hoeffding gives

$$\mathbb{P}(|\hat{p}(a, s) - p(a, s)| > \alpha) \leq 2 \exp(-2n_0\alpha^2).$$

491 If all true probabilities are at least  $p_{\min}$ , then the log-probability error obeys  $|\log \hat{p} - \log p| \leq$   
 492  $\alpha/p_{\min} + O(\alpha^2/p_{\min}^2)$  on the good event. A union bound over  $|\mathcal{A}||\mathcal{S}|$  cells yields the conservative  
 493 pilot rule

$$n_0 = O\left(\frac{\log(|\mathcal{A}||\mathcal{S}|/\delta_{\text{conf}})}{\delta_{\text{acc}}^2 p_{\min}^2}\right).$$

494 Wilson intervals and bootstrap intervals are reported in practice because pilot cell counts are small.

## 495 B Operational diagnostics for the four terms Operational diagnostics

496 The identity above becomes useful only after each term is tied to quantities logged by the deployment  
 497 protocol. The checker-facing progress process is the common object used to operationalize compe-  
 498 tence, cost, information, and allocation diagnostics. Fix a horizon  $H$  and a task-normalized checked  
 499 progress process  $G_t(a, x) \in [0, 1]$  for action  $a$  on instance  $x$ . In the main experiment, the router acts  
 500 once before deployment: after observing  $Z$ , it selects one action, and that action is used for the whole  
 501 trajectory. The success and cost diagnostics below are therefore trajectory-level quantities computed  
 502 from the step-wise verifier trace. For lower-is-better checker losses, one admissible normalization is

$$G_t(a, x) = \text{clip}_{[0,1]}\left(\frac{L_0(x) - L_t(a, x)}{|L_0(x)| + \varepsilon_L}\right),$$

503 where  $L_0(x)$  is the starting-artifact loss and  $L_t(a, x)$  is the checked loss convention at step  $t$ . Given a  
 504 success threshold  $\delta > 0$ ,

$$\tau_\delta(a, x) = \inf\{t \leq H : G_t(a, x) \geq \delta\}, \quad F_\delta(a, x) = \mathbf{1}\{\tau_\delta(a, x) = \infty\}.$$

505 We also log persistence through

$$\text{Occ}_{\delta, H}(a, x) = \frac{1}{H} \sum_{t=1}^H \mathbf{1}\{G_t(a, x) \geq \delta\}.$$

506 **Competence.** The action competence component  $p(a, s)$  is the first-success probability of action  $a$   
 507 on task mode  $s$ :

$$p(a, s) = \mathbb{P}[\tau_\delta(a, X) \leq H \mid S = s].$$

508 Thus  $G_t$  is step-wise, but the policy competence term uses the probability that the worker selected by  
 509 the router reaches the success threshold at least once before the horizon. Cumulative progress and  
 510 persistence are not substituted for  $p(a, s)$  in the identity. We log

$$\bar{G}_a(s) = \mathbb{E}\left[\frac{1}{H} \sum_{t=1}^H G_t(a, X) \mid S = s\right]$$

511 and  $\text{Occ}_{\delta, H}$  as secondary diagnostics, especially when first-hit success saturates or when partial  
 512 progress is scientifically relevant.

513 **Cost.** The action cost component  $\kappa(a, s)$  is kept separate from competence. In the main protocol  
 514 it is the normalized cost of deploying action  $a$  on mode  $s$  for the same trajectory convention used  
 515 to define  $p(a, s)$ . The routed-policy cost term averages this mode-conditional cost over the actions  
 516 selected under  $Z$ . The primary convention is cost accumulated until the first certified success or the  
 517 horizon,  $\tau_\delta \wedge H$ . We log

$$C_\delta(a, x; \xi) = \lambda^\top (\tilde{T}_{\tau_\delta \wedge H}^{\text{worker}}, \tilde{C}_{\tau_\delta \wedge H}^{\text{tok}})(a, x; \xi),$$

518 and report worker wall-clock cost as primary, with token count as a sensitivity analysis. Checker  
 519 runtime is logged for audit, but it is not part of  $\kappa(a, s)$  unless the checker protocol or checker-call  
 520 frequency differs across actions.



521 **Routing information.** The theorem term  $I(S; Z)$  measures how much a router-visible signal  $Z$   
 522 separates the predeclared task modes before deployment. This does not mean that the evaluator  
 523 already knows the best worker for each instance. The evaluator knows only the candidate mode  
 524 schema and can estimate whether  $Z_j$  makes that schema more identifiable:

$$I(S; Z_j | Z_0) = H(S | Z_0) - H(S | Z_j).$$

525 The quantity is estimated only when a finite signal representation or calibrated predictive model for  
 526 the signal records is predeclared. Otherwise the paper reports signal summaries and router-response  
 527 diagnostics instead of a Shannon mutual-information estimate. It is a descriptive information  
 528 diagnostic; its decision value is established only after action outcomes show that the modes predict  
 529 different cost–success profiles. The paired router experiment then checks whether higher-information  
 530 signals actually change the selected action and improve first-passage deployment outcomes.

531 **Allocation-summary mismatch and allocation regret.** The theorem term  $\mathbb{E}_Z \text{KL}(\pi_Z \| q_Z)$  is  
 532 defined only after the evaluator predeclares a reproducible construction of the evaluator-side mode  
 533 summary  $q_Z$ . Operationally, the router returns an action, not a mode label or distribution. When no  
 534 such  $q_Z$  construction is frozen, the experiment reports allocation regret as a separate action-frontier  
 535 diagnostic. After deployment runs, the evaluator estimates the mode-conditional frontier

$$a^*(s) \in \arg \min_{a \in \mathcal{A}} \{\log \kappa(a, s) - \log p(a, s)\},$$

536 or the statistically indistinguishable set of frontier actions. For a selected action  $a = \rho_j(x)$  and mode  
 537  $s = \sigma(x)$ , the corresponding allocation regret diagnostic is

$$r_j(x) = [\log \kappa(a, s) - \log p(a, s)] - \min_{a' \in \mathcal{A}} [\log \kappa(a', s) - \log p(a', s)].$$

538 This regret is computed retrospectively on a diagnostic split and validated on holdout instances; it  
 539 is not an oracle input to the router. High information with high allocation regret means that the  
 540 progressive signal exposed useful mode structure, but the induced action did not exploit the empirical  
 541 cost–success frontier.

542 After these diagnostics are estimated, the experiment still reports the decision-facing first-passage  
 543 deployment loss in (5) as final validation. The full-horizon audit loss in (6) is reported separately for  
 544 persistence and final-quality checks. These losses are not replacements for the theorem; they test  
 545 whether designs favored or explained by the diagnostics improve the practical deployment objective.

## 546 **C Operational-decomposition-under-progressive-signals** **Progressive-signal** 547 **decomposition**

548 The main four-term identity compares a deployed design to a prior-matched baseline. For the V3  
 549 router experiment, it is often useful to compare two progressive signal conditions for the same router.  
 550 Assume a finite task mode  $S \in \mathcal{S}$  and, for each signal condition  $j$ , the conditional mode distribution  
 551 induced by the realized signal record

$$\pi_j(s | Z_j) = \mathbb{P}[S = s | Z_j].$$

552 The fixed orchestrator  $R$  induces the routing policy  $a_R^j(Z_j) = R(Z_j)$ . If an evaluator-side mode  
 553 summary  $q_R^j(\cdot | Z_j) \in \Delta(\mathcal{S})$  has been predeclared, define

$$\mathcal{E}_R^j(s, Z_j) = \frac{\kappa(a_R^j(Z_j), s)}{p(a_R^j(Z_j), s) q_R^j(s | Z_j)}.$$

554 The log-effort gain of signal condition  $Z_j$  over  $Z_0$  is

$$\Delta_R^{\log}(j) := \mathbb{E}[\log \mathcal{E}_R^0(S, Z_0) - \log \mathcal{E}_R^j(S, Z_j)].$$

555 Expanding the cross-entropy terms gives

$$\Delta_R^{\log}(j) = C_R(j) + \Phi_R(j) + G(j) + M_R(j),$$

556 where

$$\begin{aligned}
C_R(j) &= \mathbb{E} \left[ \log \frac{\kappa(a_R^0(Z_0), S)}{\kappa(a_R^j(Z_j), S)} \right], \\
\Phi_R(j) &= \mathbb{E} \left[ \log \frac{p(a_R^j(Z_j), S)}{p(a_R^0(Z_0), S)} \right], \\
G(j) &= H(S \mid Z_0) - H(S \mid Z_j), \\
M_R(j) &= \mathbb{E} \left[ \text{KL}(\pi_0(\cdot \mid Z_0) \parallel q_R^0(\cdot \mid Z_0)) - \text{KL}(\pi_j(\cdot \mid Z_j) \parallel q_R^j(\cdot \mid Z_j)) \right].
\end{aligned}$$

557 When  $Z_j$  refines  $Z_0$ ,  $G(j) = I(S; Z_j \mid Z_0)$ . This signal-mode decomposition is theory-facing and  
558 excludes measurement cost unless such a term is added explicitly. It is a diagnostic companion to the  
559 deployment-gain estimand (8), not a replacement for it.

## 560 **D Operational-estimator-details** Full selection protocol and experimental 561 configuration

562 **Fixed-prototype panel.** The main experiment is a fixed-prototype repeated-trajectory panel rather  
563 than a population draw from  $\mu$ . Each task mode  $s$  is represented by one executable prototype

$$\chi_s = (A_s, D, \text{split}, u, \eta_s, V, B_V, H, \delta),$$

564 where  $A_s$  is the editable `train.py` template and starting model class,  $D$  is CIFAR-10, `split` is  
565 the fixed train/validation split,  $u$  is the model-initialization seed,  $\eta_s$  are starting optimization and  
566 architecture hyperparameters,  $V$  is the checker,  $B_V$  is the checker budget,  $H$  is the edit-verify  
567 horizon, and  $\delta$  is the success threshold. Panel-level aggregates use  $\hat{\pi}(s) = 1/3$ . The analyzed worker  
568 menu is  $\{\text{gpt\_5\_3\_codex}, \text{gpt\_5\_4}\}$ . The paper-facing comparison sets  $\mathcal{A} = \mathcal{M}$ : each deployable  
569 action is a single worker run under the common prompt and harness.

570 **Task-mode construction.** The paper-facing task modes are named by the starting model family in  
571 the editable artifact. *MLP-flat* (`mlp_flat`) is a fully connected classifier over flattened CIFAR-10  
572 inputs; *compact CNN* (`cnn_compact`) is a small convolutional classifier with local spatial inductive  
573 bias; and *micro ResNet* (`resnet_micro`) is a reduced residual convolutional classifier with skip  
574 connections. They are not different datasets or different workers. They are evaluator-defined variants  
575 of the same CIFAR-10 edit-verify task: the dataset, split, checker, prompt harness, horizon, and  
576 success threshold are fixed, while the initial architecture family and associated template parameters  
577 change. This choice makes the mode variable operationally meaningful for the theory, because  
578 different starting artifacts can change both first-passage competence  $p(a, s)$  and cost  $\kappa(a, s)$  for the  
579 same worker.

580 **Progressive signal records.** The router sees one of four nested pre-deployment records.  $Z_0$  contains  
581 only the task name, checker objective, worker menu, and budget.  $Z_1$  adds the initial `train.py` plus  
582 a static workload summary: starting model class, parameter count, optimizer and budget settings,  
583 dataset/checker identifiers, and parse/runtime availability.  $Z_2$  adds a low-cost checker probe of the  
584 unmodified artifact at a predeclared probe budget.  $Z_3$  adds the full unmodified-artifact baseline  
585 under the deployment checker budget. Probe records include validation loss, validation accuracy,  
586 training seconds, total seconds, optimizer steps, parameter count, parse/runtime status, and probe  
587 budget. Probe runs are outside the selected worker deployment: they are charged through  $\ell_{\text{meas}}$  and  
588 do not consume the later deployment horizon or checker budget. The evaluator-only baseline used to  
589 compute  $G_t$  is router-visible only in  $Z_3$ .

590 **Experiment accounting.** The worker frontier uses 35 trajectories per task-mode-worker cell. The  
591 selected-router validation uses the same fixed panel and reports all richer signals  $j \in \{1, 2, 3\}$  rather  
592 than selecting a single best signal on the same data. Negative controls keep the signal schema and  
593 approximate context burden of the matched real signal: shuffled probes, wrong-mode probes, and  
594 synthetic noise records are charged under the same measurement/context-cost convention as the real  
595 record they mimic. For the router audit, the prompt-only router selects `gpt_5_4` on 27/30 budget-only  
596 records, 29/30 static-workload records, 30/30 probe records, and 28/30 full-scout records.

Table 3: Frozen operational configuration for the main protocol. All values are fixed within a task mode; repeated runs vary only the agentic trajectory.

| Component                                 | Fixed value                                                                                                                                                                        |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset and split                         | CIFAR-10, 50,000 training examples and 10,000 validation examples; balanced subset, label noise 0.0, imbalance ratio 1.0; the train/validation split is fixed in the main protocol |
| Data seed                                 | Common fixed data/subset seed; alternative split seeds are reported only as robustness variants                                                                                    |
| <code>train.py</code> initialization seed | SEED=42; used for Python, NumPy, PyTorch, CUDA seeds, deterministic PyTorch mode, and the training-loader generator; alternative initialization seeds are robustness variants      |
| Training transforms                       | random crop with padding 4, random horizontal flip, CIFAR-10 normalization; validation uses normalization only                                                                     |
| Training budget per checker call          | fixed-step mode with <code>A_MAX_STEPS</code> =256; time budget 120s is a fallback, not the primary scoring convention                                                             |
| Common hyperparameters                    | <code>BATCH_SIZE</code> =64, <code>NUM_WORKERS</code> =0, Adam, learning rate $5 \cdot 10^{-4}$ , weight decay $10^{-4}$ , betas (0.9, 0.999), no LR schedule                      |
| Task mode                                 | <i>MLP-flat</i> ( <code>mip_flat</code> ), <i>compact CNN</i> ( <code>cnn_compact</code> ), or <i>micro ResNet</i> ( <code>resnet_micro</code> )                                   |
| Shared template parameters                | depth 2, base channels 12, channel multiplier 2, batch norm enabled, dropout 0.0, and hidden width 48                                                                              |
| Agent horizon                             | $H = 20$ edit-verify steps; trajectories continue to $H$ after first success to measure persistence and final quality                                                              |
| Progress normalization                    | $\varepsilon_L = 10^{-8}$ in the denominator of $G_t$                                                                                                                              |
| Primary loss weights                      | $\alpha_{\text{fail}} = 1$ ; normalized worker wall-clock is the primary scalar $C_\delta$ ; token cost is a sensitivity analysis                                                  |
| Audit loss weights                        | $\alpha_{\text{occ}} = \alpha_{\text{qual}} = 1$ , $\psi(u) = 1 - u$                                                                                                               |
| Measurement cost                          | pre-deployment probes are charged through $\ell_{\text{meas}}$ using the same wall-clock normalizer as $C_\delta$ ; context-token cost is reported as sensitivity                  |
| Router configuration                      | deterministic prompt-only router with fixed model identifier, prompt hash, parser, temperature, top-p, and seed policy recorded before evaluation                                  |

597 **Promotion rule.** A routing recommendation is promoted only when worker-frontier estimates,  
598 cost sensitivity, router shift, action-frontier regret, paired gain after measurement cost, and negative  
599 controls agree. If diagnostics improve but paired deployment gain does not, the report treats the result  
600 as a mechanism failure rather than a useful router.

601 **Trajectory record.** Each run stores the worker alias, resolved model identifier, API date, split,  
602 task family, task mode, seed, maximum horizon  $H$ , signal condition, prompt hash, router model  
603 identifier, router temperature/top-p, router seed policy, parser version, router output, selected worker,  
604 and cost convention. Each step stores the prompt-visible state, proposed `train.py` artifact, parse  
605 status, checker result, validation loss, validation accuracy, relative improvement, incremental worker  
606 wall time, token usage, and checker runtime for audit. Parser failures are logged as unsuccessful  
607 routing or deployment records under the predeclared parser-failure policy.

608 **Success and time-to-verification.** For trajectory  $i$ ,  $\tau_i$  is the first step at which the relative improve-  
609 ment threshold is crossed; censored failures have  $\tau_i = \infty$ . For each action-mode cell,

$$\hat{F}_a(s, h) = \frac{1}{n_{a,s}} \sum_i \mathbf{1}\{\tau_i \leq h\}, \quad \hat{S}_a(s, h) = 1 - \hat{F}_a(s, h).$$

610 The empirical hazard is

$$\hat{\lambda}_a(s, h) = \frac{\sum_i \mathbf{1}\{\tau_i = h\}}{\sum_i \mathbf{1}\{\tau_i \geq h\}}.$$

611 The raw first-success estimate is  $\hat{p}(a, s) = \hat{F}_a(s, H)$ . For log-effort scores, the protocol uses the  
612 predeclared smoothed estimate

$$\tilde{p}(a, s) = \frac{k_{a,s} + 1/2}{n_{a,s} + 1},$$

where  $k_{a,s}$  is the number of first successes in the cell. This prevents zero-success pilot cells from making log scores infinite; raw success counts are still reported.

**Cost.** The primary cost is cumulative worker wall-clock time to  $\tau \wedge H$ . Secondary worker-side cost is cumulative token count. Checker runtime is logged as a protocol audit field and excluded from action-cost comparisons unless the checker protocol differs across actions. All plots that support model recommendations must be reproducible under at least worker wall-clock and token-count conventions. When a worker fails before producing a parseable artifact, its cost is still counted and the step is marked unsuccessful. The scalar estimator matched to Assumption 1 is the cell mean

$$\hat{\kappa}(a, s) = \frac{1}{n_{a,s}} \sum_{i: S_i = s} C_{\delta,i}(a),$$

or its fixed-prototype analogue  $\hat{\kappa}_{\chi_s}(a)$  in Section 5.

**Information, allocation summaries, and regret.** For each progressive signal  $Z_j$ , a Shannon information estimate is reported only if the protocol predeclares a finite signal representation or calibrated predictive model for estimating  $\pi_j(\cdot | Z_j)$  on held-out instances. Otherwise the report uses qualitative signal summaries and router-response diagnostics without calling them estimates of  $I(S; Z_j)$ . The KL mode-summary divergence term is reported only if the evaluator also freezes a reproducible construction of  $\hat{q}_R^j(\cdot | Z_{j,i})$  before seeing evaluation outcomes. For the deterministic router, the empirical panel shift rate is

$$\widehat{\text{Shift}}_R(j) = \sum_{s \in \mathcal{S}} \hat{\pi}(s) \mathbf{1}\{\rho_j(\chi_s) \neq \rho_0(\chi_s)\}.$$

In the main fixed-prototype protocol, the primary operational diagnostic is allocation regret,

$$\hat{r}_j(x_i) = [\log \hat{\kappa}(a_{j,i}, s_i) - \log \tilde{p}(a_{j,i}, s_i)] - \min_{a \in \mathcal{A}} [\log \hat{\kappa}(a, s_i) - \log \tilde{p}(a, s_i)],$$

where  $a_{j,i} = R(Z_j(x_i))$  and  $s_i = \sigma(x_i)$ .

**Uncertainty.** Success probabilities use Wilson intervals. Aggregated objectives, paired gains, and calibration diagnostics use bootstrap or randomization tests at the sampling unit declared for the study. If the main protocol remains a fixed-prototype repeated-run panel, paired randomization tests over repeated trajectories are descriptive for that panel and should not be interpreted as population-level inference. If independent task instances are added, uncertainty is reported at the task-instance level.

## E Additional diagnostic plots

This appendix collects auxiliary diagnostics generated from the experimental logs. The threshold, trajectory, and cost diagnostics below use the same analyzed workers, gpt\_5\_3\_codex and gpt\_5\_4. These appendix plots use the same 35-trajectory-per-cell panel as the main analysis. The pooled appendix plots are robustness and mechanism diagnostics for the same accounting quantities reported in the main text.

### E.1 Pooled threshold and trajectory diagnostics

### E.2 Pooled cost diagnostics

Threshold sensitivity, two mature workers pooled common support (n=210)

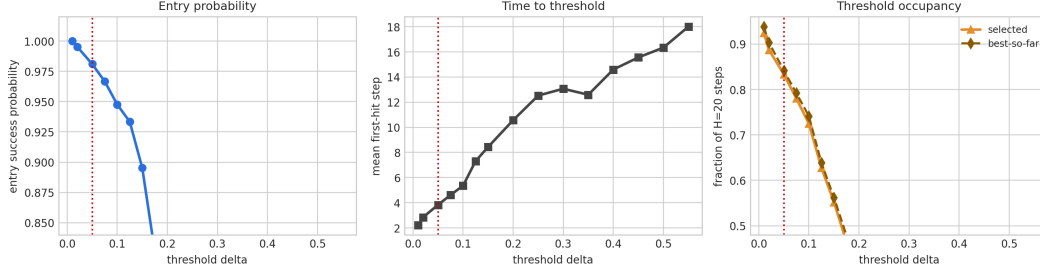


Figure 4: Zoomed threshold sensitivity for the two mature workers on the 35-run pooled panel. The red line marks the operating threshold  $\delta = 0.05$ . At this threshold, entry success is near saturated while mean first-hit time remains low, which makes  $\delta = 0.05$  a pragmatic operating point for studying time-to-certified-solution. The sharp drop at larger thresholds shows why the exact threshold is not an innocuous plotting choice.

Extended threshold sensitivity by worker, pooled common support (n=210)

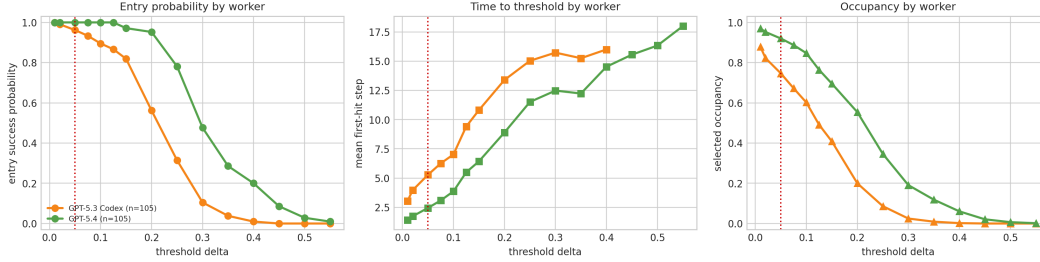


Figure 5: Threshold sensitivity by worker on the 35-run pooled panel. gpt\_5\_4 remains stronger at higher thresholds, while gpt\_5\_3\_codex degrades earlier; this supports the competence term in Theorem 1.

Extended threshold sensitivity, two mature workers pooled common support (n=210)

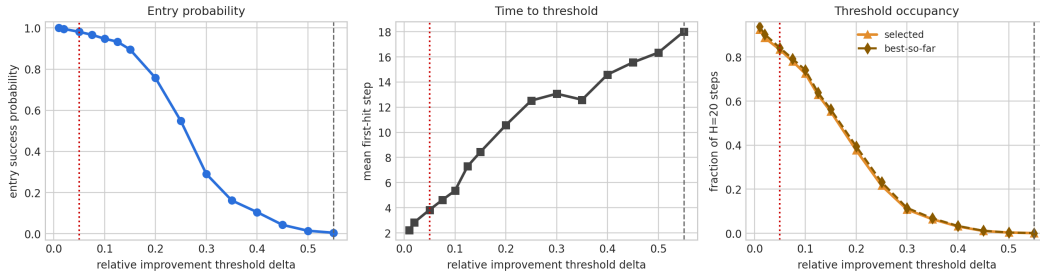


Figure 6: Extended threshold sensitivity for the two-worker pooled panel. The full threshold range shows that high-threshold success is rare even when  $\delta = 0.05$  is nearly saturated. This is why the paper reports first-passage curves, occupancy, and final quality rather than relying on a single binary success rate.

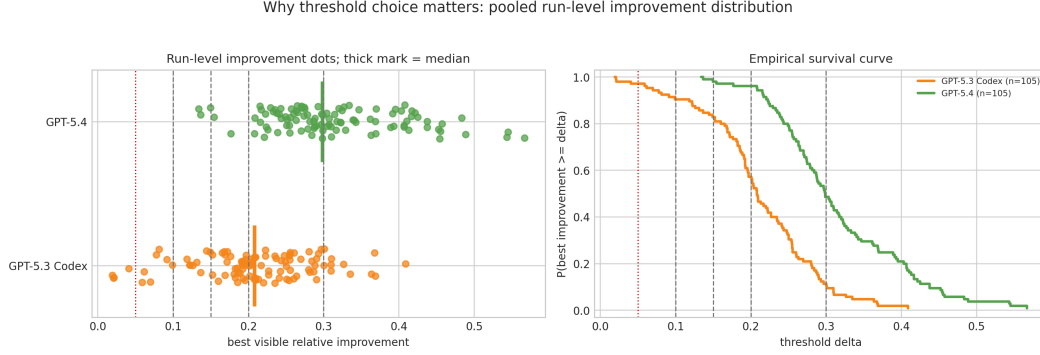


Figure 7: Run-level improvement distribution for the two mature workers. Dots show final best-visible relative improvement and thick ticks mark worker medians. The survival plot makes the threshold choice explicit: near  $\delta = 0.05$ , most runs succeed, but the ordering and separation change at higher thresholds. This diagnostic supports the paper’s claim that final quality, first-pass success, and deployment loss are distinct empirical objects.

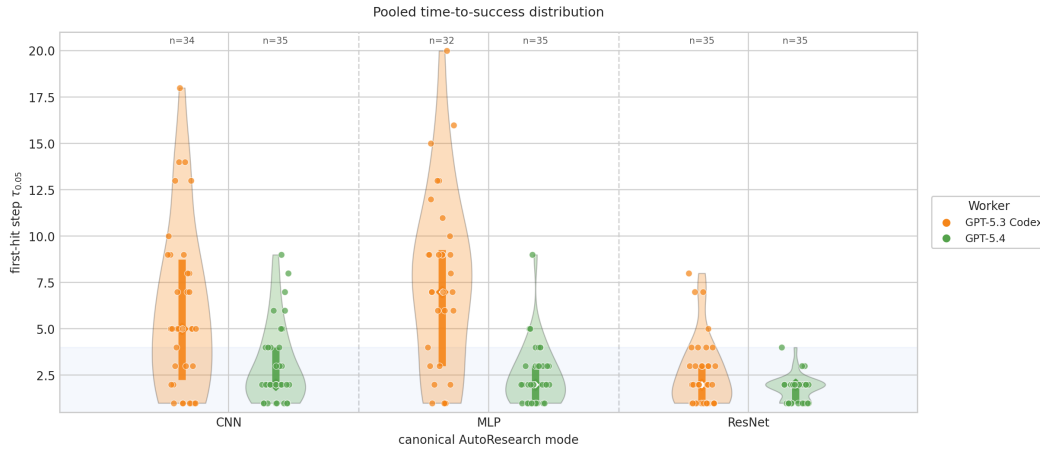


Figure 8: Pooled time-to-success distribution by task mode and worker. The violin plots expose mode-conditional heterogeneity: gpt\_5\_4 typically hits the threshold quickly, while gpt\_5\_3\_codex has broader first-hit times, especially on MLP. This is the empirical object behind the cost and competence terms of the four-term decomposition.

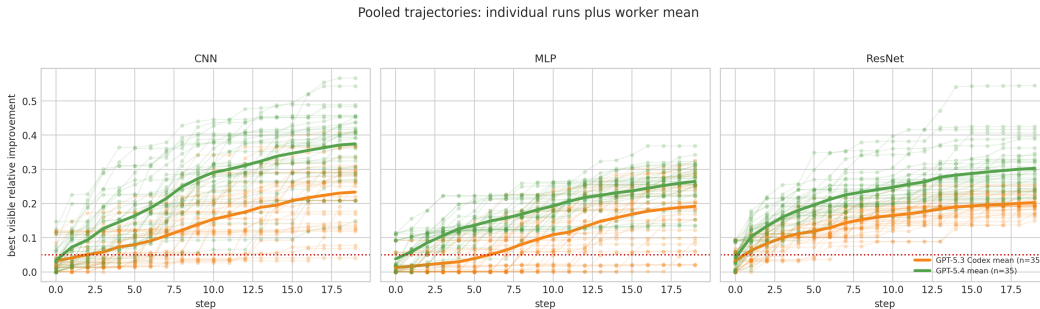


Figure 9: Best-so-far improvement trajectories. The mean trajectories show that gpt\_5\_4 improves earlier and reaches higher final relative improvement in the pooled panel, but the deployment accounting in the main text shows that this does not automatically imply lower deployment loss after cost is charged. This contrast is the main empirical reason to separate worker competence from deployment value.



## NeurIPS Paper Checklist

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and introduction state the paper’s scope: verifiable agentic systems, a packed latent-mode assumption, decomposition-guided selection, and a controlled fixed-prototype experimental protocol. The discussion section separates theory-facing and deployment-facing claims.

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Section 7 discusses the packed-mode assumption, small live-agent sample sizes, finite design menus, oracle mode labels, and current cost-measurement limitations.

### 3. Theory assumptions and proofs

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Justification: Section 5 describes the AutoResearch task instantiation, task modes, model menu, router-visible signals, horizons, metrics, and evaluation protocol; Appendix D specifies the required logs, estimators, and run-metadata fields.

### 5. Open access to data and code

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

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## 14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

742 Justification: The paper does not involve crowdsourcing or research with human subjects.

743 **15. Institutional review board (IRB) approvals or equivalent for research with human**

744 **subjects**

745 Question: Does the paper describe potential risks incurred by study participants, whether

746 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)

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748 institution) were obtained?

749 Answer: [N/A]

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753 Question: Does the paper describe the usage of LLMs if it is an important, original, or

754 non-standard component of the core methods in this research? Note that if the LLM is used

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756 scientific rigor, or originality of the research, declaration is not required.

757 Answer: [Yes]

758 Justification: LLMs are central to the evaluated agent designs. Section 5 describes the

759 three-worker menu, fixed prompt-only router, sequential signal protocol, model-identifier

760 logging, and routing-output requirements.

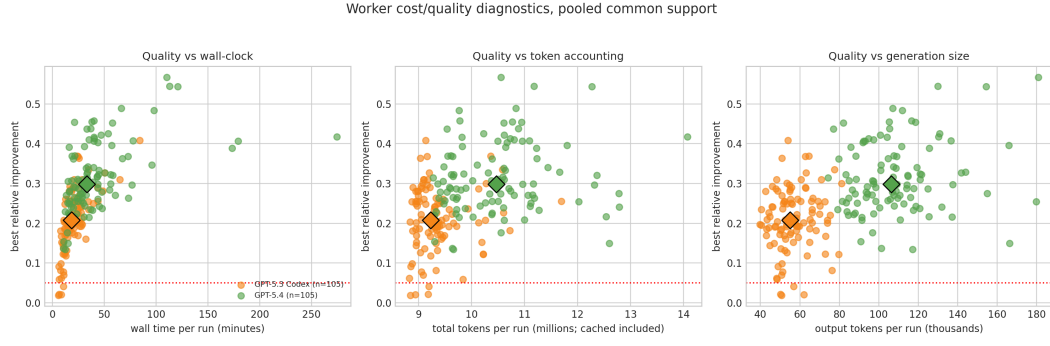


Figure 10: Worker cost–quality diagnostics. Each point is a pooled common-support run, and diamonds mark worker medians. The plots show that better final improvement is accompanied by substantial variation in wall-clock time, total token accounting, and generated-token volume. This supports the paper’s budget-aware framing: worker choice cannot be reduced to final improvement alone.

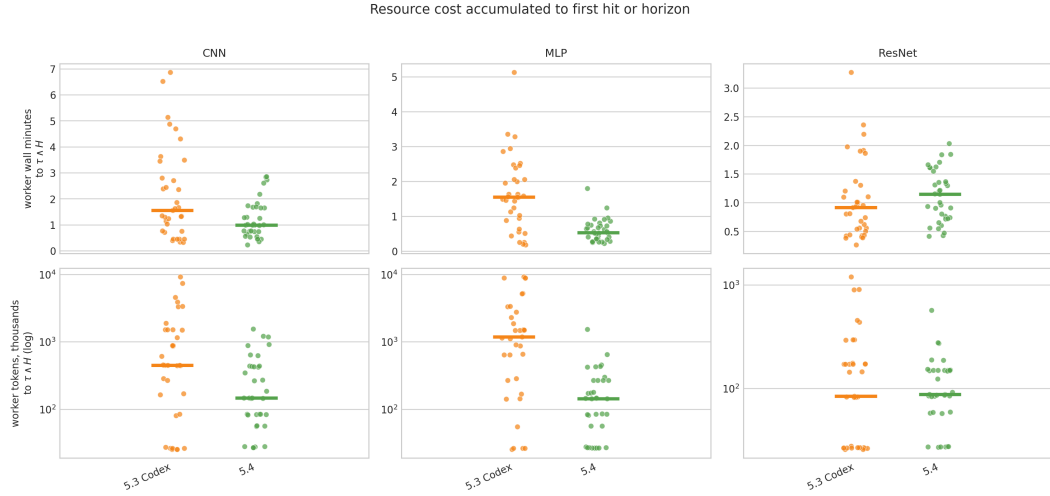


Figure 11: Resource cost accumulated to first hit or horizon. The top row reports wall-clock minutes and the bottom row reports token accounting on a log scale. Mode-conditional cost spreads are large, which is why the retry-crossover theorem is treated as a cost-sensitive design rule rather than a generic endorsement of weaker workers.

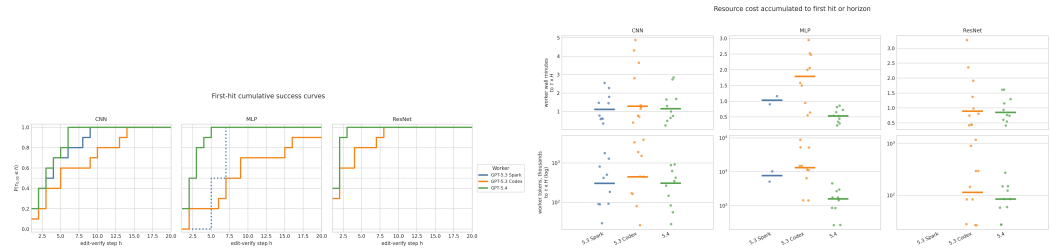


Figure 12: First-passage diagnostics. Left: first-hit CDFs by mode. Right: cost-to-threshold by mode and worker. These are the direct empirical objects behind the retry and time-to-certified-solution interpretation.

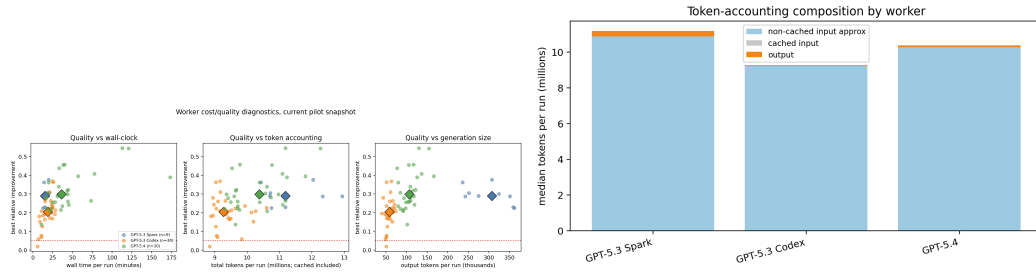


Figure 13: Worker-side cost and accounting diagnostics. These plots check whether conclusions are driven by wall-clock cost, token composition, or quality differences.

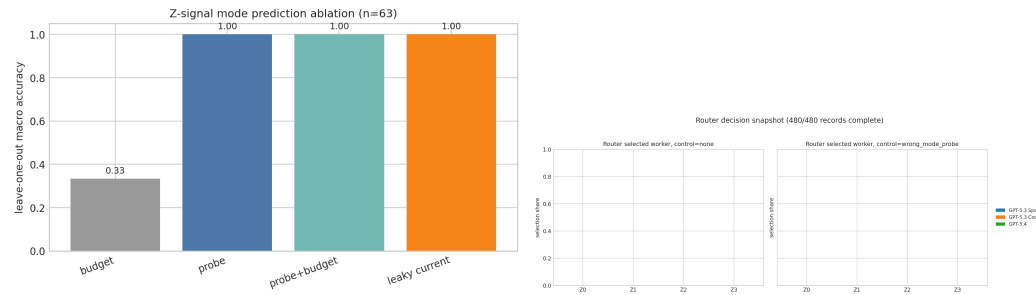


Figure 14: Router-facing diagnostics. The signal ablation tests whether progressively richer records expose task-mode structure; the router-selection snapshot checks whether the prompt-only router changes action distribution under those records.

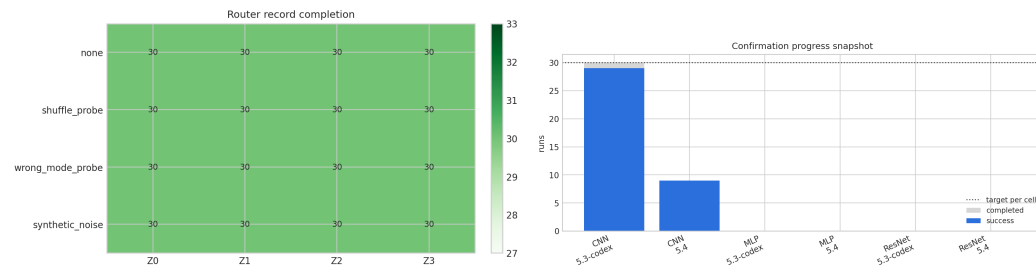


Figure 15: Run-coverage diagnostics for router records and holdout confirmation runs. These plots document which analyses are part of the promoted two-worker comparison and which are exploratory.